illumina®

Illumina Single Cell 3′ RNA Prep, T10

Product Documentation

ILLUMINA PROPRIETARY
Document # 200063182 v01
August 2025

For Research Use Only. Not for use in diagnostic procedures.

This document and its contents are proprietary to Illumina, Inc. and its affiliates ("Illumina"), and are intended solely for the contractual use of its customer in connection with the use of the product(s) described herein and for no other purpose. This document and its contents shall not be used or distributed for any other purpose and/or otherwise communicated, disclosed, or reproduced in any way whatsoever without the prior written consent of Illumina. Illumina does not convey any license under its patent, trademark, copyright, or common-law rights nor similar rights of any third parties by this document.

The instructions in this document must be strictly and explicitly followed by qualified and properly trained personnel in order to ensure the proper and safe use of the product(s) described herein. All of the contents of this document must be fully read and understood prior to using such product(s).

FAILURE TO COMPLETELY READ AND EXPLICITLY FOLLOW ALL OF THE INSTRUCTIONS CONTAINED HEREIN MAY RESULT IN DAMAGE TO THE PRODUCT(S), INJURY TO PERSONS, INCLUDING TO USERS OR OTHERS, AND DAMAGE TO OTHER PROPERTY, AND WILL VOID ANY WARRANTY APPLICABLE TO THE PRODUCT(S).

ILLUMINA DOES NOT ASSUME ANY LIABILITY ARISING OUT OF THE IMPROPER USE OF THE PRODUCT(S) DESCRIBED HEREIN (INCLUDING PARTS THEREOF OR SOFTWARE).

© 2025 Illumina, Inc. All rights reserved.

All trademarks are the property of Illumina, Inc. or their respective owners. For specific trademark information, refer to www.illumina.com/company/legal.html.

Table of Contents

Introduction 1
Consumables & Equipment 2
Product Contents 2
User-Supplied Consumables & Equipment 5
Equipment Preparation 9
PIPseq Vortexer Operation 9
Dry Bath Operation 9
Sample Preparation 12
Protocol 13
Tips and Techniques 15
Prepare for Protocol 17
Prepare Cell Suspension 18
Prepare Nuclei Suspension 20
Capture and Lysis 22
Prepare for Protocol 25
Isolate mRNA 25
Normalize PIP Volume 28
Synthesize cDNA 29
Prepare for Protocol 30
Wash PIPs for cDNA Amplification 31
Amplify cDNA 31
Prepare for Protocol 33
Isolate cDNA from PIPs 33
Clean Up cDNA 34
Assess cDNA Quality 36
Prepare for Protocol 37
Clean Up QC Product 38
Check QC Product 39
Prepare for Protocol 40
Fragmentation, End Repair, & A-Tailing 41
Ligate Adapters 42
Clean Up Ligated Fragments 42
Amplify Library 44

For Research Use Only. Not for use in diagnostic procedures.

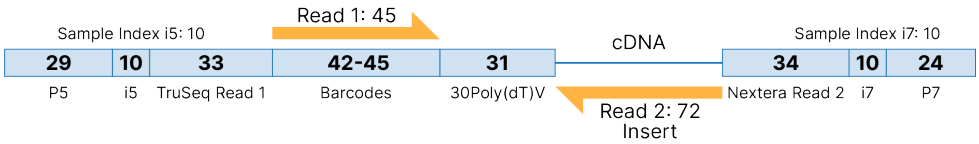
Prepare for Protocol 45
Clean Up Library 46
Check Library 47
Dilute Libraries to the Starting Concentration 48
Resources & References 49
Revision History 49

For Research Use Only. Not for use in diagnostic procedures.

Introduction

The Illumina Single Cell 3′ RNA Prep, T10 kit uses particle-templated instant partitions (PIPs) to combine single cell or nuclei suspensions with barcoded beads to produce sequencing-ready libraries in less than two days. The assay supports eight individual reactions, in which up to 10,000 cells or nuclei can be profiled per reaction.

Illumina Single Cell 3′ RNA Prep, T10 kit libraries are composed of standard Illumina paired-end constructs that begin with P5 and end with P7. Read 1 contains the barcode information while Read 2 contains the gene expression information. These libraries are dual-indexed with 10-base i5 and i7 indexes. Read 1 length must be ≥ 45 bases and the recommended Read 2 length ≥ 72 bases.



Illumina Single Cell 3′ RNA Prep libraries can be pooled for sequencing, taking into account the differences in cell number and read depth requirements. Sequencing depth varies based on your application needs but it is recommended to start with a depth of 20,000 reads per input cell. Samples must have distinct index combinations to avoid failures in sample demultiplexing. Refer to *Index Adapters Pooling Guide* for discussion of appropriate color balance combinations for the selected sequencing platform. After the Illumina Single Cell 3′ RNA Prep libraries are quantified and normalized, dilute and prepare the libraries as recommended in Denature and Dilute Protocol Generator for the selected Illumina sequencing system.

For Research Use Only. Not for use in diagnostic procedures.

Consumables & Equipment

The Illumina Single Cell 3′ RNA Prep, T10 protocol requires the following Illumina-supplied and user-supplied consumables and equipment.

The protocol has been optimized and validated using the items listed. Comparable performance is not guaranteed when using alternate consumables and equipment.

Make sure that you have the required consumables and equipment before starting the protocol.

Product Contents

The protocol requires one capture and barcoding kit, one library preparation kit, and one equipment kit. If you are purchasing the Illumina Single Cell Vortexer and Accessories kit, a third-party dry bath is required. Refer to *User-Supplied Consumables & Equipment* on page 5 for more information on the third-party dry bath and accessories supported with this workflow.

Component	Kit Name	Illumina Catalog #
Library prep with capture and barcoding	Illumina Single Cell 3′ RNA Prep, T10 (8 samples, 10,000 cells/sample)	20135691
Single cell capture and barcoding only	Illumina Single Cell 3′ RNA Capture, T10 (8 samples, 10,000 cells/sample)	20132791
Library prep only	Illumina Single Cell Library Prep (8 reactions)	20132789
[Optional] Index adapters	Illumina Single Cell Unique Dual Indexes (96 Indexes, 96 Samples)	20132788
Starter equipment	Illumina Single Cell Prep Starter Equipment	20132796
	Illumina Single Cell Vortexer and Accessories	20145920

Illumina Single Cell 3′ RNA Capture, T10 Contents

Table 1 Illumina Single Cell 3′ RNA Capture, T10, Box 1, Store at 15°C to 30°C

Quantity	Reagent	Description
1	PAR	Partitioning Reagent
1	CLB3	Chemical Lysis Buffer 3
1	DPR	Departitioning Reagent

Table 2 Illumina Single Cell 3′ RNA Capture, T10, Box 2, Store at 2°C to 8°C

Quantity	Reagent	Description
1	IPB2	Illumina Purification Beads 2
1	BBF	Breaking Buffer
1	PWB1	Wash Buffer
1	CEB	cDNA Extraction Buffer

Table 3 Illumina Single Cell 3′ RNA Capture, T10, Box 3, Store at -25°C to -15°C

Quantity	Reagent	Description
1	NSB	Nuclei Suspension Buffer
1	RTEM	RT Enzyme Mix
2	RTAM	RT Additive Mix
1	PMM4	4X PCR Master Mix
1	WTAP	WTA Primer
1	RNI	RNase Inhibitor (40 U/µl)
1	CSB	Cell Suspension Buffer
1	BSA	25% BSA

Table 4 Illumina Single Cell 3′ RNA Capture, T10, Box 4, Store at -85°C to -65°C

Quantity	Reagent	Description
8	PIP	T10 PIPs
4	TSO1	Template-Switch Oligo

Table 5 Illumina Single Cell 3′ RNA Capture, T10, Accessories Bag

Quantity	Component
16	1 ml syringe
16	G22 blunt-bottom syringe needle

Illumina Single Cell Library Prep Contents

Table 6 Illumina Single Cell Library Prep, Box 1, Store at 2°C to 8°C

Quantity	Reagent	Description
1	IPB2	Illumina Purification Beads 2
1	NFW	Nuclease-Free Water

For Research Use Only. Not for use in diagnostic procedures.

Table 7 Illumina Single Cell Library Prep, Box 2, Store at -25°C to -15°C

Quantity	Reagent	Description
1	LPB	Library Prep Buffer
1	LPE	Library Prep Enzymes
1	LPMA	Library Prep Mix A
1	LAM	Library Adapter Mix
1	PMM4	4X PCR Master Mix
1	UDI8	Illumina Single Cell Prep UDI Mix Strip

Illumina Single Cell Prep Starter Equipment Kit

Quantity	Component Name	Used for Kit size
1	PIPseq Vortex Mixer	All
1	PIPseq rotating vortex assembly for 5 ml tubes	T100
1	PIPseq rotating vortex assembly for 1.5 ml tubes	T20
1	PIPseq rotating vortex assembly for 0.5 ml tubes	T2 and T10
2	Rotating vortex base assembly	All
1	User manual for PIPseq Vortex Mixer	All
1	US Power Supplies for Vortex Mixer	All
1	Vortexer gap cover	All
2	PIPseq Combination 1.5 ml and 0.5 ml tube stand, blue	T2,T10,T20
1	PIPseq 8-tube stand, for 0.2 ml tubes, black	T2
1	PIPseq guide rack, red	T2
2	International Plug Converter	All
1	PIPseq Dry Bath 5 ml block	T100
1	PIPseq Dry Bath 1.5 ml block	T20
1	PIPseq Dry Bath 0.5 ml block	T2 and T10
1	US Power Supplies for PIPseq Dry Bath	All
1	User Manual for PIPseq Dry Bath	All

Quantity	Component Name	Used for Kit size
1	PIPseq Dry Bath with heated lid, which includes the following materials: • 2.5 mm Allen key wrench, silver • PIPseq Dry Bath Stylus • Replacement tips for PIPseq Dry Bath Stylus • 5x20 mm 5A 250V Fast blow fuse for PIPseq Dry Bath • Dry block removal tool, white	All

Illumina Single Cell Vortexer and Accessories

Quantity	Component Name	Used for Kit size
1	PIPseq Vortex Mixer	All
1	PIPseq rotating vortex assembly for 5 ml tubes	T100
1	PIPseq rotating vortex assembly for 1.5 ml tubes	T20
1	PIPseq rotating vortex assembly for 0.5 ml tubes	T2 and T10
2	Rotating vortex base assembly	All
1	User manual for PIPseq Vortex Mixer	All
1	US Power Supplies for Vortex Mixer	All
1	Vortexer gap cover	All
2	PIPseq Combination 1.5 ml and 0.5 ml tube stand, blue	T2,T10,T20
1	PIPseq 8-tube stand, for 0.2 ml tubes, black	T2
1	PIPseq guide rack, red	T2
2	International Plug Converter	All

User-Supplied Consumables & Equipment

The following sections provide information on the required user-supplied consumables and equipment.

The specific brands of plastic consumables have been validated for stability of PIP emulsions. Substituting these materials can adversely affect performance.

For Research Use Only. Not for use in diagnostic procedures.

Consumables

Consumable	Supplier
0.2 ml PCR 8-tube strips and caps, emulsion safe	One of the following suppliers: • Greiner Bio-One, catalog # 673210 (tubes) and catalog # 373270 (caps) • USA Scientific, catalog # 1402-1700 • Millipore Sigma, catalog # BR781327
0.5 ml Safe-Lock PCR Clean tubes, emulsion safe	One of the following suppliers: • VWR, catalog # 62111-750 • Fisher Scientific, catalog # 05-402-93 • Eppendorf, catalog # 022363719
1.5 ml Safe-Lock PCR Clean tubes, emulsion safe	One of the following suppliers: • VWR, catalog # 62111-752 • Fisher Scientific, catalog # 05-402-94 • Eppendorf, catalog # 022363212
[Fresh cells] 5 ml polypropylene centrifuge tubes, sterile	Fisher Scientific, catalog # 13-864-407
[Cryopreserved cells] 15 ml polypropylene centrifuge tubes, sterile	Corning, catalog # 430790 or general lab supplier
Absolute ethanol, molecular biology grade	General lab supplier
[Isolated nuclei input] BSA, molecular biology grade	General lab supplier
Cell counting materials (eg, AO/PI)	General lab supplier
Nuclease-free water	Thermo Fisher, catalog # AM9937 or general lab supplier
Nuclease-free 1X low EDTA TE Buffer (IDTE), pH 8.0, 10 mM Tris, 0.1 mM EDTA	IDT, catalog # 11-05-01-09 or general lab supplier
Pipette tips, filtered, low retention, sterilized, 20 µl	One of the following suppliers: • Rainin, catalog # 30389226 • Filtrous, catalog # PTF-LS-0020 • VWR, catalog # 76322-528
Pipette tips, filtered, low retention, sterilized, 200 µl	One of the following suppliers: • Rainin, catalog # 30389240 • Filtrous, catalog # PTF-LS-0200 • VWR, catalog # 76322-150

Consumable	Supplier
Pipette tips, filtered, low retention, sterilized, 1000 µl	One of the following suppliers: • Rainin, catalog # 30389213 • Filtrous, catalog # PTF-LS-1000 • VWR, catalog # 76322-154
Pipette tips, wide bore, filtered, low retention, sterilized, 200 µl	Rainin, catalog # 30389241
Pipette tips, wide bore, filtered, low retention, sterilized, 1000 µl	Rainin, catalog # 30389218
PCR tube strips and caps, or PCR reaction plates (if you have a compatible magnetic bead rack)	General lab supplier
Quantification consumables, depending on system: • Bioanalyzer High Sensitivity DNA kit • Tapestation • HS-D5000 for cDNA • HS-D1000 for NGS library	One of the following suppliers, depending on system: • Agilent, catalog # 5067-4626 • Agilent, catalog # HS-D5000 5067-5592 and 5067-5593 • Agilent, catalog # HS-D1000 5067-5584 and 5067-5585
Qubit 1X dsDNA High Sensitivity Assay Kit	Thermo Fisher, catalog # Q33230 or Q33231
Qubit Assay Tubes	Thermo Fisher, catalog # Q32856
[Isolated nuclei input] RNase Inhibitor	One of the following suppliers: • Thermo Fisher, catalog # AM2694 • Sigma, catalog # RNAINH-RO • Solis Biodyne, catalog # 06-26-4000U • Watchmaker Genomics, catalog # 7K0088
[Optional] Cell Strainers	General lab supplier

Equipment

Equipment	Supplier
0.2 ml/1.5 ml Magnetic Separation Combo Rack	Permagen, MSR1224B or equivalent from alternative supplier
Benchtop centrifuge with swinging bucket rotor, suitable for 5/15 ml tubes	General lab supplier, recommend Thermo Scientific ST40

Equipment	Supplier
Benchtop mini microcentrifuge 2000 × g, suitable for 0.5/1.5 ml tubes and PCR tube strips.	Thermo Fisher, catalog # 75004061 or general lab supplier
[Optional] Dry bath* The following components must be ordered separately: • Heated lid • 0.5 ml block and transfer rack	Eppendorf, catalog # 5383000027 Eppendorf, catalog # 5308000003 Eppendorf, catalog # 5361000031
Hemocytometer or automated cell counter	General lab supplier
Ice bucket or cold blocks, suitable for 0.2/0.5 ml PCR tubes and 1.5 ml microcentrifuge tubes	General lab supplier
Micropipettes, 1–1000 µl capabilities (Rainin LTS-tip compatible pipettes are required if you are using the recommended Rainin pipette tips)	General lab supplier
Quantification system, either of the following: • Bioanalyzer 2100 • Tapestation 4200	One of the following suppliers, depending on system: • Agilent, catalog # G2940CA • Agilent, catalog # G2991BA
Qubit Fluorometer	Thermo Fisher
Thermal cycler	General lab supplier
Vortexer	General lab supplier

*The dry bath and its associated components are only required if you purchased the Illumina Single Cell Vortexer and Accessories starter equipment kit.

For Research Use Only. Not for use in diagnostic procedures.

Equipment Preparation

This section provides instructions for preparing equipment for use with the Illumina Single Cell 3′ RNA Prep, T10 kit. For additional dry bath and vortex mixer documentation, including detailed installation, and operation instructions, refer to the Illumina Single Cell 3′ RNA Prep pages on the Illumina support site. Make sure that you are familiar with the operation of all equipment before starting the protocol.

Stater equipment documentation:

- *PIPseq Vortex Mixer Operating Instructions (document # 200064269)*
- *PIPseq Vortexer Adapter Installation (document # 200064267)*
- *Dry Bath Incubator with Heated Lid Operating Instructions (document # 200064268)*

PIPseq Vortexer Operation

The protocol requires a vortexer with a 0.5 ml vortexer head. Make sure that the appropriate head is installed with the thumbscrew inserted into the front of the vortexer.

Review the process for rotating the vortex head from horizontal to vertical tube position. The process should be completed in less than 30 seconds.



Horizontal tube position



Vertical tube position

Dry Bath Operation

The protocol requires a dry bath with 0.5 ml tube block. Make sure that the appropriate block is installed.

The dry bath steps specified in this protocol are specific to the PIPseq dry bath sold as part of the Illumina Single Cell Prep Starter Equipment Kit. If you are using a user-supplied Eppendorf dry bath, you must manually program the lysis and reverse transcription protocols. Refer to Eppendorf's documentation for more detailed information on how to operate their products.

For Research Use Only. Not for use in diagnostic procedures.

Dry Bath Control

- Lid temperature is set manually and is independent from the dry bath programs. Understand how to configure lid mode and temperature before starting the protocol.
  - The dry bath interface displays **LidOn** if the lid temperature control is not active. Select **LidOn** to activate the lid temperature control. When the lid temperature control is active, the interface displays **LidOff**.
  - For nuclei lysis and reverse transcription, use the specific temperature control mode to set the lid temperature to 105°C. When set to this mode, the lid temperature box shows the set value (for example, 105.0).
  - For cell lysis, use the offset temperature control mode to set the lid temperature to 5°C higher than the heat block. When set to this mode, the lid temperature box shows the temperature offset (for example +5.0).
  - To change the lid mode, select **Edit**, and then select **LidMode** to toggle between 105°C and +5.0, and then select **Save/Return**.
- The lid temperature only climbs to 105°C during incubation temperatures of 35°C or higher.
- The dry bath is preprogrammed with nuclei lysis, cell lysis, and reverse transcription protocols.
- Preheat and hold settings are depicted as 00:00.

Dry Bath Programs

The following protocols are preprogrammed on the dry bath:

Program A (Cell Lysis)

- Preheat 25°C
- 25°C for 15 minutes
- 37°C for 45 minutes
- 25°C for 10 minutes
- Hold at 20°C

Program B (Nuclei Lysis)

- Preheat 66°C
- 66°C for 45 minutes
- 25°C for 10 minutes
- Hold at 20°C

Program C (Reverse Transcription)

- Preheat 25°C
- 25°C for 30 minutes
- 42°C for 90 minutes

- 85°C for 10 minutes
- Hold at 4°C

For Research Use Only. Not for use in diagnostic procedures.

Sample Preparation

The Illumina Single Cell 3′ RNA Prep protocol requires a suspension of high-quality single cells or nuclei as input derived from cell cultures, dissociated tissues, cell sorting, or other isolation methods.

The following guidelines apply to sample preparation:

- Minimize the presence of dead cells or aggregates to ensure the highest quality data.
- Minimize the amount of time that it takes for sample preparation through capture and lysis, and add RNase inhibitor at 0.4–1 U/µl final concentration to buffers being used in lengthy steps (dissociation, enrichment, sorting collection tubes, or sample suspension buffers).
- Use higher RNase inhibitor concentrations for challenging sample types.
- When processing many samples together at once, Illumina recommends processing sample preparations in batches through capture and lysis to prevent mRNA degradation.

Depending on your input type, refer to the *Prepare Cell Suspension* on page 18 or *Prepare Nuclei Suspension* on page 20 for sample preparation instructions. Start capture and lysis as soon as possible after diluting samples to the target concentration.

Working with Fixed Cells

Fixed cells generated according to *DSP-Methanol Fixation for Cells Protocol Instructions* are compatible with this workflow. If cells are in media containing FBS or BSA, complete a 1 ml wash step using Cell Suspension Buffer, then resuspend in Cell Suspension Buffer that does not include additional BSA or RNase inhibitor before starting the fixation protocol. This fixative yields comparable performance as fresh cells. Fixed cells can be stored for up to seven days at -25°C to -15°C. Alternative fixation approaches (eg, formaldehyde) have not yet been validated and are not recommended.

Working with Fixed Nuclei

Fixed nuclei are compatible with this workflow. Before starting the fixation protocol nuclei must be resuspended in Nuclei Suspension Buffer (NSB) without BSA or RNase inhibitor. Refer to *DSP-Methanol Fixation for Nuclei Protocol Instructions* for instructions on fixing isolated nuclei. You are strongly advised to substitute NSB working solution for the wash steps in the alternative nuclei fixation protocol (at least 1 ml of NSB working solution without BSA or RNase inhibitor in up to two wash steps and the final resuspension before fixation). Fixed nuclei can be stored for up to two months at -25°C to -15°C.

For Research Use Only. Not for use in diagnostic procedures.

Protocol

This section provides step-by-step instructions on how to use the kit to generate sequence-ready libraries from input sample. Safe stopping points are indicated at the end of applicable steps.

- Before proceeding, confirm the kit contents and make sure that you have the necessary consumables and equipment. Refer to *Consumables & Equipment* on page 2.
- Follow the protocol in the order described using the specified parameters. Unless a safe stopping point is specified, move immediately to the next step.

Illumina Single Cell 3′ RNA Workflow



For Research Use Only. Not for use in diagnostic procedures.

Tips and Techniques

Working with RNA

Before executing this protocol, become familiar with working with RNA. The following section provides general guidelines for working with RNA, but the guidelines are not all-inclusive.

- Due to the ubiquitous presence of RNases, RNA is susceptible to degradation. RNases are robust nucleases specific to RNA. Unlike DNases, they are not easily denatured or inactivated.
- Maintain reagents, pipettes, and a work space specifically for working with RNA. An enclosed space such as a PCR hood is recommended. Wipe down pipettes and the working space regularly with a laboratory alcohol cleaning solution (70% alcohol). Periodically (one time a week to one time a month), clean surfaces with a 10% sodium hypochlorite solution. Alternatively, commercial solutions are available to inactivate RNases.
- Only use consumables and reagents that are nuclease-free and dedicated for RNA use. Handle reagents carefully to avoid RNase contamination.
- The most common external RNase contamination comes from the skin. Handle any item used for RNA work with gloves to maintain nuclease-free surfaces. Wear a clean laboratory coat or gown to prevent shedding of skin or hair in and around the RNA work surface.
- Dust and bacteria are also sources of RNases. Keep surfaces and items free of dust and work with RNA away from bacterial processes.
- Treat refrigerator and freezer handles as RNase-free surfaces and only open them with gloved hands.
- Exercise all standard laboratory best practices such as always wearing PPE, not touching rims of open tubes, and not reusing pipette tips. Seal consumable bags and reagent tubes tightly when not in use.

Handling Beads

- Do not overdry beads. Proceed to the next step when the magnetic beads have air dried to the point of only a slight gloss remaining, but before they show cracks. This can take as few as 2 minutes depending on the tube size and humidity level.

Working with PCR Products

Before executing this protocol, become familiar with the process of unidirectional PCR workflow and managing PCR amplicon products. Complete all steps in the Illumina Single Cell 3′ RNA workflow in a pre-PCR workspace until after the thermal cycling step during Amplify Library. Complete all remaining steps in the post-PCR workspace. One exception is the magnetic bead purification following Assess cDNA Quality, which also must be completed in the post-PCR workspace. The following section provides general guidelines for working with PCR products, but the guidelines are not all-inclusive.

- PCR by design creates many amplicons, copies of the input material. This strength of PCR also comes with the risk of amplicon contamination of workspaces and materials.
- It is best practice to maintain two separate workspaces for pre and post-PCR work. Ideally these spaces are physically divided and each have a dedicated set of equipment and materials. All work before a PCR amplification is to be performed in the pre-PCR workspace. The sample is then transferred to the post-PCR workspace for further processing. No material from the post-PCR workspace should ever come back into the pre-PCR workspace.
- For some reagents and consumables, multiple sets may be required for pre-PCR and post-PCR workspace use to avoid carryover contamination.
- If it is not possible to maintain two separate workspaces for pre and post-PCR work, it is still highly recommended to keep two separate sets of consumable materials and to maintain strict workplace sterility. Make sure all benches and equipment are regularly cleaned with a bleach solution or a commercial DNA degrader to reduce contaminating amplicons and always wear clean gloves.

Centrifuge Steps

- Complete all centrifuge steps in the Illumina Single Cell 3′ RNA workflow after sample preparation with a benchtop minifuge using speeds of 2000 × g.
- Always use the power button to turn the benchtop minifuge off for a gradual stop to maintain well packed beads. Do not apply standard brakes, like opening the lid.
- Use 0.5 ml tube inserts in the benchtop minifuge to prevent tube opening.

Thermal Cycler Lid Pressure

Follow the manufacturer's instructions for using PCR tube strips with your brand of thermal cycler. Thermal cyclers with a lot of lid pressure can cause the 8-tube strip to collapse during PCR amplification steps when only one is used. Use empty tube strips as braces surrounding both sides of the 8-tube strip to help balance the lid pressure for brands that do not include plastic frame supports.

PCR 8-Tube Strip Lids

When vortexing 8-tube strips, apply pressure to the lid to maintain a tight seal. Replace the lids on the 8-tube strip after each dry bath or thermal cycler program using the 105°C lid setting to prevent lid damage.

Cell Loading

This protocol describes addition of 17,000 cells into each reaction, resulting in recovery of > 10,000 cells and a multiplet rate of < 8%. The optimal input cell concentration is 3,400 live cells/µl when adding 5 µl cells plus 40 units RNase inhibitor into each PIP tube. If it is not possible to concentrate cells to 3,400 cells/µl, a larger volume can be input as long as the total loading volume including RNase inhibitor is kept ≤ 6 µl. It is important to note that increased cell input increases the observed multiplet rate.

Prepare for Protocol

This preparation is required to perform the protocol steps leading up to the next safe stopping point.

1. Remove sample from storage.
2. Remove third-party reagents from storage.

   Table 8 Third-Party Reagents

Reagent	Storage	Thaw Instructions
RNase Inhibitor*	Varies for third-party reagents	Keep on ice. Refer to the manufacturer instructions as needed.
[Cryopreserved cells] Thawing media	Varies for third-party reagents	Refer to the manufacturer instructions as needed.

   * For cell and nuclei suspensions, use user-supplied RNase Inhibitor.

3. Remove the reagents from the box and thaw as follows.

   Table 9 15°C to 30°C Storage

Reagent	Box Name	Thaw Instructions
PAR	Illumina Single Cell 3′ RNA Capture, T10, Box 1	Use at room temperature.
CLB3	Illumina Single Cell 3′ RNA Capture, T10, Box 1	Use at room temperature.

   Table 10 -25°C to -15°C Storage

Reagent	Box Name	Thaw Instructions
[Cell preparation] CSB	Illumina Single Cell 3′ RNA Capture, T10, Box 3	For cryopreserved cells—Warm to 37°C in a water bath. For fresh cells—Keep cold.
[Nuclei preparation] NSB	Illumina Single Cell 3′ RNA Capture, T10, Box 3	Use at room temperature.
RNI*	Illumina Single Cell 3′ RNA Capture, T10, Box 3	Keep on ice.
25% BSA	Illumina Single Cell 3′ RNA Capture, T10, Box 3	Thaw on ice.

   * For capture and lysis, use Illumina-supplied RNI.

For Research Use Only. Not for use in diagnostic procedures.

Table 11 -85°C to -65°C Storage

Reagent	Box Name	Thaw Instructions
T10 PIP	Illumina Single Cell 3′ RNA Capture, T10, Box 4	Thaw one tube per sample on ice. Stable on ice for up to 5 hours.

Prepare Cell Suspension

Use this protocol to prepare single-cell suspensions from mammalian fresh or cryopreserved cells. Adjustments to centrifuge speed and times might be required for other cell types. If you are using your own cell preparation protocol, the final cell dilution step must be performed with the Cell Suspension Buffer provided in the kit.

If you are preparing single cell suspensions from tissues or fragile cell types, additional dissociation or cell handling steps might be required. These steps are not provided in the following protocol.

Consumables

- Cell Suspension Buffer
- **[Cryopreserved cells]** Thawing media
- **[Fresh cells]** 5 ml conical tube
- **[Cryopreserved cells]** 15 ml conical tube
- 1000 µl wide bore, low retention, pipette tips
- Cell counting materials
- **[Optional]** Cell strainer

Preparation

- Thawing media—Warm to 37°C in a water bath.

Procedure

Thaw Cryopreserved Cells

If you are starting from fresh cells, proceed to *Prepare Cell Suspension* on page 19.

1. Place the cryopreserved cell vial and an aliquot of Cell Suspension Buffer in a water bath set to 37°C and let stand in the water bath until a small amount of ice remains (~1 minute).
2. Remove the vial from the water bath, decontaminate the outside with 70% isopropyl alcohol, and then move it into the biosafety cabinet.
   The remaining ice thaws over the next 30–60 seconds at room temperature.

For Research Use Only. Not for use in diagnostic procedures.

3. Use a wide-bore 1000 µl pipette to slowly pipette to mix, and then transfer the cell suspension from the cell vial to the conical tube.
4. Add thawing media to the cell suspension as follows.
   a. Slowly add 2 ml thawing media (~30 seconds).
   b. Add the remaining 7 ml thawing media.
   c. Invert five times to mix.

Prepare Cell Suspension

1. Centrifuge cells at 200 × g for 5 minutes at room temperature to pellet.
   Use a swinging bucket rotor for pelleting to minimize cell shearing and loss. Optimize minimum speed and time required for pelleting your specific cell type. For example, you can use up to 350 g × 5 minutes for smaller cell types.
2. Without disturbing the pellet, remove and discard the supernatant.
   If it is not possible to leave less than 100 µl supernatant remaining, increase the wash volume using the following table as guidance.

   The wash step removes reagents inhibitory to the assay from the cell suspension (eg, FBS, BSA > 1%, high salt or calcium, carryover organic solvents). Failure to remove inhibitory reagents can cause cDNA QC failure.

3. Add the appropriate volume of Cell Suspension Buffer, and then place the remaining Cell Suspension Buffer on ice.

Remaining Supernatant Volume (µl)	Cell Suspension Buffer Volume (ml)
< 100 µl	1
100–200 µl	2

4. Using a wide-bore 1000 µl pipette tip, slowly pipette 5 times to mix.
5. Centrifuge at 200 × g for 3 minutes at room temperature to pellet.
   Optimize minimum speed and time required for pelleting your specific cell type. For example, you can use up to 350 g × 5 minutes for smaller cell types.
6. Without disturbing the pellet, remove and discard the supernatant.
7. Using a standard-bore pipette tip, add 200–400 µl chilled Cell Suspension Buffer, and then gently pipette 10–15 to mix.
   If clumps are visible, increase the amount of force used during pipette mixing. Use the minimum force and number of pipette mixes necessary to create a homogeneous cell suspension.
8. **[Optional]** To remove debris and dead cells, pass the cell suspension through a strainer as follows.

   Filtering can cause cell stress and up to 30% cell loss. It is recommended only for cell types prone to forming aggregates.

For Research Use Only. Not for use in diagnostic procedures.

a. Use a wide-bore pipette tip to withdraw 200 µl cell suspension.
b. Slowly dispense the cell suspension through the cell strainer into a fresh 1.5 ml tube.
Use minimal force to push the cell suspension through the strainer to avoid unwanted debris in the cell suspension.
c. Repeat a and b until all cell suspension has been passed through the cell strainer.

9. Use a cell counter to determine cell concentration.
Cells should have > 90% viability to proceed. If necessary, perform additional wash steps to remove debris and improve viability. Always perform a final count.
10. Using a wide-bore low retention pipette tip, prepare 3,400 live cells/µl suspension in Cell Suspension Buffer to achieve an input volume of 5 µl (17,000 live cells).
If it is not possible to concentrate cells to 3,400 cells/µl, a larger volume can be input as long as the total loading volume including 40 units of RNase inhibitor is ≤ 6 µl.
11. Repeat the cell count described in step 9 to confirm the final concentration in the prepared suspension. Adjust concentration and repeat count as necessary.
12. After the required concentration is achieved, place the cells on ice and proceed immediately to *Capture and Lysis* on page 22.

Prepare Nuclei Suspension

Illumina recommends that you prepare nuclei suspensions using the Illumina Single Cell Nuclei Isolation Kit to ensure consistent, high-quality nuclei across frozen mammalian tissues. Refer to the *Illumina Single Cell Nuclei Isolation Kit Product Documentation* for instructions.

If using an alternative nuclei isolation protocol, use the provided NSB to prepare NSB working solution (refer to *Prepare NSB Working Solution* on page 21) for the final nuclei suspension. Substitute NSB working solution for the wash steps in the alternative nuclei isolation protocol (at least 1 ml of NSB working solution in up to two wash steps). This kit only includes enough BSA for the final nuclei suspension, so make sure to add user-supplied BSA (1% final concentration) and RNase inhibitor (0.8 U/µl final concentration) if needed for the wash steps.

Alternative Nuclei Isolation Protocols

- When using alternative nuclei isolation protocols, it is necessary to include 0.8 U/µl RNase inhibitor final concentration and 1X protease inhibitor into the nuclei extraction buffer/nuclei lysis buffer during tissue lysis.
- After nuclei extraction is done, centrifuge nuclei suspension (500 × g for 5 minutes at 4°C) to pellet and aspirate supernatant. Use a 200 µl pipette to carefully remove as much supernatant as possible without disturbing the pellet.

> Using a fixed angle centrifuge for pelleting nuclei for wash steps can result in excessive sample loss. Use tubes compatible with swinging bucket rotor centrifuges to ensure a flat pellet.

For Research Use Only. Not for use in diagnostic procedures.

- Add 1 ml of NSB working solution, mix well and spin at 500 × g for 5 minutes at 4°C.
- Aspirate supernatant without disturbing the pellet and resuspend nuclei using NSB working solution.
- If unable to leave less than 100 µl supernatant remaining, increase the wash volume to 2 ml.

> The wash step removes reagents inhibitory to the assay from the nuclei suspension (eg, FBS, BSA > 1%, high salt or calcium, carryover organic solvents). Failure to remove inhibitory reagents can cause cDNA QC failure.

Refer to *Prepare NSB Working Solution* on page 21 for instruction on preparing Nuclei Suspension Buffer for various applications. Refer to the Illumina Single Cell Nuclei Isolation Kit product documentation for recommendations on the resuspension volume per mg tissue for various tissue types. It is not recommended to process more than 50 mg of tissue per sample at once to achieve optimal lysis.

After preparing the nuclei suspension, place samples on ice and proceed immediately to *Capture and Lysis* on page 22. Do not freeze isolated nuclei. If not proceeding directly to capture and lysis, it is recommended to fix nuclei to prevent mRNA degradation. Refer to *Prepare Nuclei Suspension* on page 20 for details.

Prepare NSB Working Solution

To prepare 1 ml of NSB working solution, combine the volumes from the following table. Prepare sufficient volume for the final resuspension with NSB working solution per sample. Exact volumes required might be sample size and type specific to achieve optimal nuclei concentration.

Refer to *DSP-Methanol Fixation for Nuclei Protocol Instructions* for the volume to use for the final suspension before fixation.

Component Name	NSB Working Solution Volumes for Fresh Nuclei (µl)	NSB Working Solution Volumes for Fixed Nuclei (µl)
Nuclei Suspension Buffer	167	167
Nuclease-Free Water	773	833
25% BSA	40	Not applicable
RNase Inhibitor (40 U/µl)	20*	Not applicable
Total	1000	1000

* Alternatively, you can use 40 µl RNase inhibitor if provided at 20 U/µl and 753 µl nuclease-free water. Refer to RNase inhibitor recommendations in *User-Supplied Consumables & Equipment* on page 5.

Capture and Lysis

Consumables

- CLB3 (Chemical Lysis Buffer 3)
- PAR (Partitioning Reagent)
- RNI (RNase Inhibitor)
- T10 PIP
- 0.5 ml Safe-Lock PCR Clean tubes (emulsion safe)
- 200 µl standard bore, low retention, pipette tips
- 200 µl wide bore, low retention, pipette tips
- 1 ml syringe
- G22 blunt bottom syringe needle

Preparation

1. Prepare the following consumables:
   - CLB3—If crystals are observed, place on the provided dry bath set to 25°C to equilibrate the tubes to room temperature. Vortex the tubes for 10 seconds, and then centrifuge briefly.
   - T10 PIP—Centrifuge for 5 seconds. Keep on ice.
2. Label one 0.5 ml Safe-Lock PCR Clean tube per reaction CLB3.
3. Confirm the vortexer program is set to 3000 rpm for 2 minutes.
4. Depending on your sample type, select one of the following options to set the lid temperature:
   - For nuclei lysis, select **Prog.B**, and then manually set the lid temperature to 105°C.
   - For cell lysis, select **Prog.A**, and then manually set the lid temperature to +5.0.

   Refer to *Dry Bath Operation* on page 9 to review program parameters.
5. Select **Run**, and then select **Lid On** to preheat the dry bath to the hold temperature for the applicable lysis program.

   The dry bath steps specified in this protocol are specific to the PIPseq dry bath sold as part of the Illumina Single Cell Prep Starter Equipment Kit. If you are using a user-supplied Eppendorf dry bath, you must manually program the protocols. Refer to Eppendorf's documentation for more detailed information on how to operate their products.

Procedure

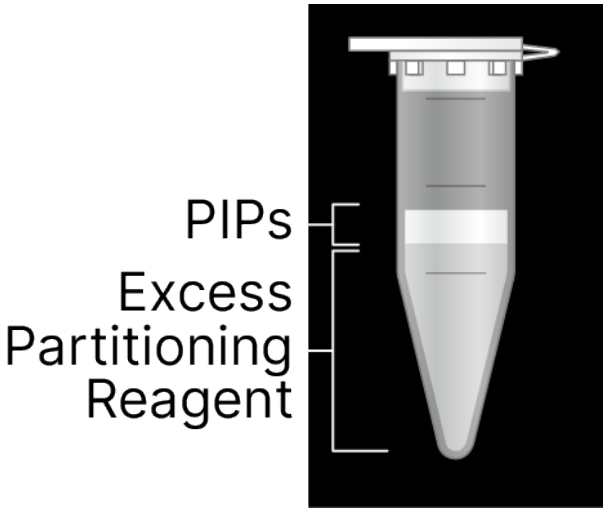
1. Using a wide-bore 200 µl pipette tip, slowly pipette the cell or nuclei suspension 10 times to mix.

For Research Use Only. Not for use in diagnostic procedures.

2. Add 5 µl cell or nuclei suspension at 3,400 cells/µl (17,000 total) directly into the PIPs layer in each PIP tube.
3. Add 40 units RNase Inhibitor directly into the PIPs layer.
   Dispense slowly to avoid forming bubbles.
4. Using a standard-bore 200 µl pipette tip, slowly pipette 10 times at the 75 µl stroke to mix.
   When removing the pipette, dispense along the side wall of the tube to avoid excessive foam.

   Generating excessive foam requires additional mixing steps that can cause stress on the cells and lead to poor assay results.

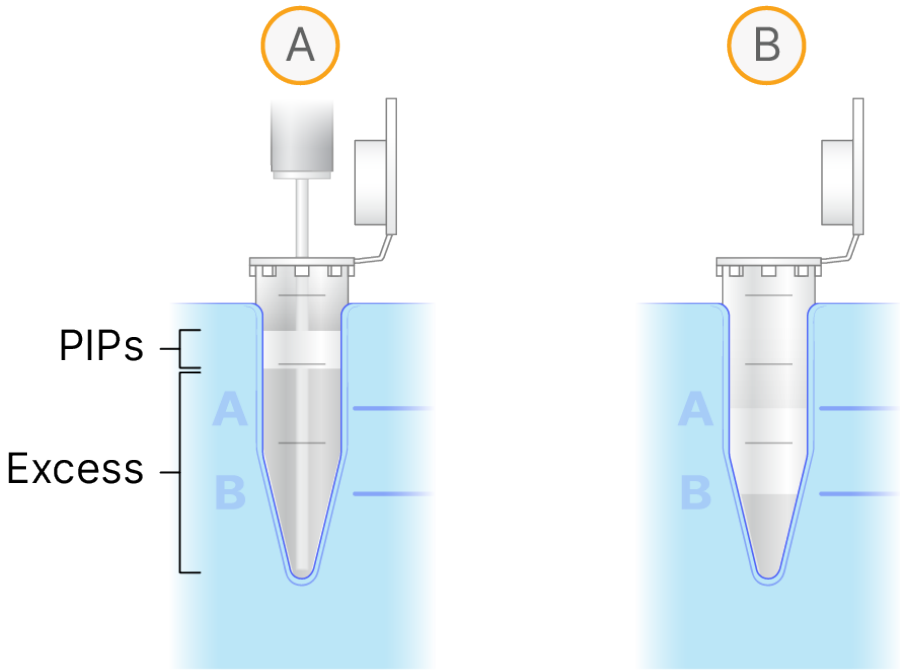
   To mix multiple PIP tubes at once, use a pipette tip on every other channel of a 200 µl multichannel pipette in an alternating configuration.
5. If excessive foam forms, centrifuge briefly and then slowly pipette 10 times at the 75 µl stroke.
6. Using a 1000 µl pipette tip, add 320 µl PAR down the side wall of the PIP tube.
7. Tightly cap tubes and place in the vortex adapter head in the horizontal tube position. Ensure the tubes are fully inserted into the adapter.
8. Vortex at 3000 rpm and manually stop the program after 15 seconds.
9. Rotate the vortex adapter into the vertical tube position, and then vortex at 3000 rpm for 2 minutes.
10. Place tubes in the 0.5 ml side of the combination tube stand and wait until the emulsion separates into two distinct layers (~30 seconds).
    The top layer contains PIPs and cells. The bottom layer contains PAR.



11. Using a G22 blunt bottom syringe needle on a 1 ml syringe, remove and discard the PAR waste (bottom phase) as follows.
    a. Slowly lower the syringe tip through the PIPs emulsion (figure A) to the bottom of the tube.
    b. Wait 5 seconds.

c. Aspirate the bottom phase until the bottom of the PIPs emulsion is even with the B stand mark (figure B). Be careful not to aspirate any emulsion.

d. While removing the tip, wipe it on the inside of the tube.



Do not place the PIP tubes on ice for the remainder of the Capture and Lysis steps to avoid precipitation of the lysis reagents.

12. Combine the following volumes per reaction in the CLB3 tube to prepare the Chemical Lysis Emulsion:
    - CLB3 (90 µl)
    - PAR (270 µl)

    Do not prepare the Chemical Lysis Emulsion in bulk. Prepare each sample aliquot individually.
13. Vortex the Chemical Lysis Emulsion for 10 seconds, and then use a 1000 µl low retention tip to immediately transfer the entire volume of one CLB3 tube to each PIP tube.
    Mixing CLB3 with PAR before transferring is critical to form the emulsion.
14. Invert 10–15 times to mix.
15. Place on the preheated dry bath. Select **Skip**, and then select **Yes** to run the applicable lysis program.
    If you are immediately proceeding to *Isolate mRNA* on page 25, prepare reagents while the lysis program is running.

SAFE STOPPING POINT

If you are stopping, leave the tube on the dry bath at 20°C for up to 96 hours. Do not refrigerate or freeze samples.

Prepare for Protocol

This preparation is required to perform the protocol steps leading up to the next safe stopping point.

1. Remove the reagents from the box and thaw as follows.

Table 12 15°C to 30°C Storage

Reagent	Box Name	Thaw Instructions
DPR	Illumina Single Cell 3′ RNA Capture, T10, Box 1	Use at room temperature.

Table 13 2°C to 8°C Storage

Reagent	Box Name	Thaw Instructions
BBF	Illumina Single Cell 3′ RNA Capture, T10, Box 2	Let stand for 30 minutes to bring to room temperature.
PWB1	Illumina Single Cell 3′ RNA Capture, T10, Box 2	Keep on ice.

Table 14 -25°C to -15°C Storage

Reagent	Box Name	Thaw Instructions
RTAM	Illumina Single Cell 3′ RNA Capture, T10, Box 3	Thaw on ice.
RTEM	Illumina Single Cell 3′ RNA Capture, T10, Box 3	Thaw on ice.

Table 15 -85°C to -65°C Storage

Reagent	Box Name	Thaw Instructions
TSO1	Illumina Single Cell 3′ RNA Capture, T10, Box 4	Thaw one tube per two samples on ice.

Isolate mRNA

Consumables

- BBF (Breaking Buffer)
- DPR (Departitioning Reagent)

- PWB1 (Wash Buffer)
- 0.5 ml Safe-Lock PCR Clean tubes (emulsion safe)
- 1.5 ml Safe-Lock PCR Clean tubes (emulsion safe)
- 1 ml syringe
- G22 blunt bottom syringe tip
- Low-retention pipette tips

Preparation

1. Prepare PWB1 as follows.
   a. For each sample, label a 1.5 ml tube Wash 1.
   b. Aliquot 1 ml PWB1 to each tube.
   c. Place the Wash 1 tubes on ice.

Procedure

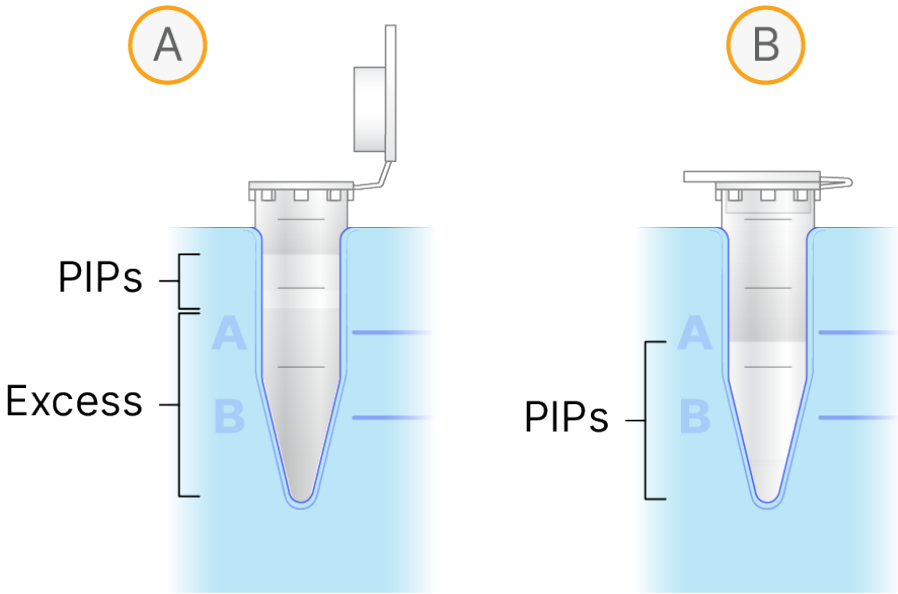
Break Emulsions

1. Remove PIP tubes from the dry bath and place into the tube stand.

   If condensation appears on the tubes, invert the sample 3 times and let the emulsion settle (~1 minute). Do not centrifuge the emulsion.

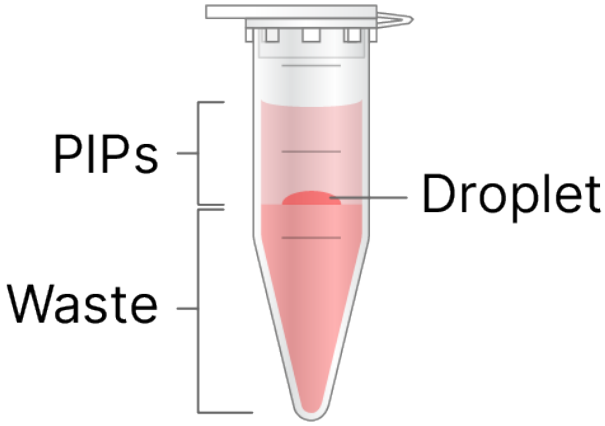
2. Using a G22 blunt bottom syringe needle on a 1 ml syringe, remove and discard the remaining PAR as follows.
   a. Slowly lower the syringe tip through the emulsion (figure A) to the bottom of the tube.
   b. Wait 5 seconds.
   c. Aspirate the bottom phase until the top of the emulsion is below the A marker on the stand (figure B).
   Do not discard any of the PIP layer.
   d. While removing the tip, wipe it on the inside of the tube.
   e. Save the syringe and needle from this step for future steps. If you are processing multiple samples, label the syringe with the sample name.

For Research Use Only. Not for use in diagnostic procedures.



Remove as much of the bottom phase as possible. To minimize sample loss, follow best practices for working with PIPs.

3. Add 300 µl BBF (clear) to each sample, along the side wall of the tube.
4. Add 100 µl DPR (pink) to each sample, along the side wall of the tube.
5. Securely close the tube, then fully invert the tube 10–20 times to break the emulsion.
6. Centrifuge for 10 seconds.
7. Ensure that the emulsion is completely broken by visually confirming a distinct interface between the pink bottom phase and the upper homogeneous aqueous layer containing the PIPs.
8. Check for any inhomogeneity in the opaque upper phase, which is a sign of an unbroken emulsion.
9. If an unbroken emulsion is observed, add 10 µl additional DPR and repeat steps 5 and 6.
10. If the unbroken emulsion still remains, add 50 µl additional DPR and repeat steps 5 and 6.
11. Using the syringe, remove all of the pink-colored bottom phase. If a red droplet is observed, remove it with the pink-colored bottom phase.



For Research Use Only. Not for use in diagnostic procedures.

12. Centrifuge for 10 seconds.
13. Using the syringe, remove residual pink-colored bottom phase.

    It is critical to ensure all of the pink bottom phase is removed. It is acceptable to remove a small amount of the clear aqueous to ensure complete removal. If there are unbroken emulsions or pink waste remaining, it can inhibit reverse transcription and cause lower cDNA yield and fragment sizes.

14. Place the tubes on ice, and then proceed immediately to the next step.

    Do not allow the PIP tubes to freeze. This negatively impacts assay performance.

Wash PIPs

1. Using a 200 µl low-retention tip, slowly transfer the PIPs from the PIP tube to the chilled Wash 1 tube.
   When transferring PIPs, ensure no PIPs are left in the pipette tip by mixing up and down at least 3 times in the Wash 1 tube.
2. Briefly centrifuge the PIP tube to collect remaining PIPs.
3. Transfer all remaining PIPs from the PIP tube to the corresponding Wash 1 tube.
4. Place the Wash 1 tubes into the stand.
5. Vortex mix by holding the stand horizontally on a flat-head vortex mixer for ~3 seconds.
6. Centrifuge for 1 minute.
7. Without disturbing the pellet, remove supernatant to the 3 volume mark on the stand.

   The pellet is translucent at this step. Removing excess volume can result in loss of PIPs.

8. Perform an additional wash as follows.
   a. Add 1200 µl 1X PWB1 to each tube.
   b. Vortex mix by holding the stand horizontally on a flat-head vortex mixer for ~3 seconds.
   c. Centrifuge the tubes for 1 minute, and then return tubes to the stand.
   d. Without disturbing the pellet, remove supernatant to the 3 volume mark on the stand.
9. Repeat steps 8 a–d to wash PIPs a **third** time. Use 1000 µl 1X PWB1.
10. Repeat steps 8 a–d to wash PIPs a **fourth** time. Use 1000 µl 1X PWB1.

Normalize PIP Volume

This step normalizes sample concentrations for cDNA synthesis.

Consumables

- 0.5 ml Safe-Lock PCR Clean tubes (emulsion safe)

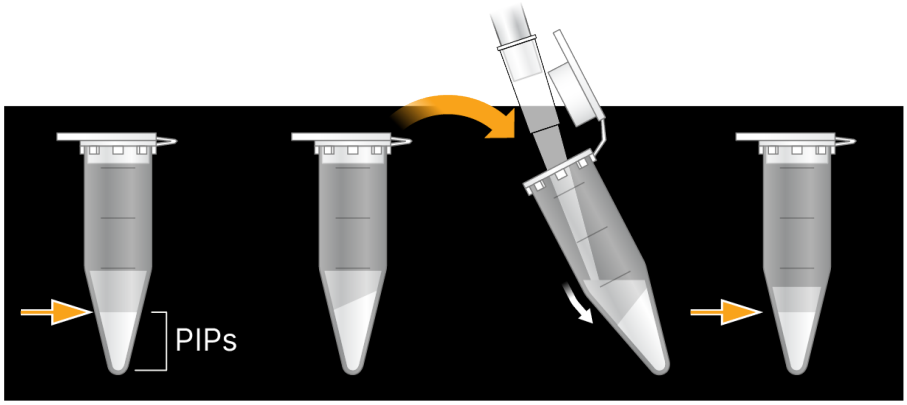
For Research Use Only. Not for use in diagnostic procedures.

Preparation

1. Use a permanent marker to mark the 0.1 ml volume engraving on the tube.

Procedure

1. Transfer the entire volume (~200 µl) of each PIP mixture to a marked 0.5 ml Safe-Lock PCR Clean tube.
2. Centrifuge for 1 minute.
   Process two reactions at a time during centrifuge and supernatant removal steps to ensure optimal PIP pelleting.
3. Keeping the tube at an angle, use a 200 µl pipette to immediately remove the supernatant down to the 0.1 ml mark on the tube.
   Do not disrupt packed PIPs.



4. Place tubes on ice and proceed immediately to cDNA synthesis.

Synthesize cDNA

Consumables

- RTAM (RT Additive Mix)
- RTEM (RT Enzyme Mix)
- TSO1 (Template-Switch Oligo)

Preparation

1. Prepare the following consumables:
   - RTAM—Pipette to mix.
   - RTEM—Pipette to mix.
2. Select **Prog.C** and then manually set the lid temperature to 105°C on the dry bath.

Refer to *Dry Bath Operation* on page 9 to review program parameters.

3. Select **Run**, and then select **Lid On** to preheat the dry bath to 25°C for the reverse transcription program.

> The dry bath steps specified in this protocol are specific to the PIPseq dry bath sold as part of the Illumina Single Cell Prep Starter Equipment Kit. If you are using a user-supplied Eppendorf dry bath, you must manually program the protocols. Refer to Eppendorf's documentation for more detailed information on how to operate their products.

Procedure

1. Combine the following volumes to create RT Master Mix.

Reagent	Volume Per Reaction (µl)	4.4X Reaction Master Mix (µl)	8.8X Reaction Master Mix (µl)
RTAM	80	352	704
TSO1	8	35	70
RTEM	12	53	106
Total	100	440	880

2. Pipette to mix.
3. Centrifuge briefly to bring liquid to the bottom of the tube.
4. Add 100 µl RT Master Mix to each PIP tube.
5. Flick to disperse the pellet.
6. Pulse vortex for 5 seconds to mix, and then centrifuge for < 1 second.
   If PIPs are pelleted in the centrifuge, repeat vortexing and centrifuge steps.
7. Place on the preheated dry bath. Select **Skip**, and then select **Yes** to run the cDNA Synthesis program, starting with incubation at 25°C for 30 minutes.

SAFE STOPPING POINT

If you are stopping, leave the samples in the dry bath at 4°C or store samples at 2°C to 8°C overnight.

Prepare for Protocol

This preparation is required to perform the protocol steps leading up to the next safe stopping point.

1. Remove the reagents from the box and thaw as follows.

Table 16 2°C to 8°C Storage

Reagent	Box Name	Thaw Instructions
PWB1	Illumina Single Cell 3′ RNA Capture, T10, Box 2	Keep on ice.

Table 17 -25°C to -15°C Storage

Reagent	Box Name	Thaw Instructions
PMM4	Illumina Single Cell 3′ RNA Capture, T10, Box 3	Keep on ice.
WTAP	Illumina Single Cell 3′ RNA Capture, T10, Box 3	Thaw on ice.

Wash PIPs for cDNA Amplification

Consumables

- PWB1 (Wash Buffer)
- Nuclease-free water (customer provided)

Procedure

1. Combine the following volumes to prepare 0.5X PWB1. Multiply each volume by the number of samples.
   - PWB1 (625 µl)
   - Nuclease-free water (625 µl)

   These volumes produce 1250 µl 0.5X PWB1 per sample.
2. Centrifuge the PIP tubes for 1 minute.
3. Without disturbing the pellet, remove and discard 130 µl supernatant.
4. Wash PIPs as follows.
   a. Add 450 µl 0.5X PWB1.
   b. Vortex for 5 seconds.
   c. Centrifuge for 1 minute.
   d. Without disturbing the pellet, remove and discard supernatant to the A mark on the tube holder.
5. Repeat steps 4 a–d to wash PIPs a **second** time. Use 350 µl 0.5X PWB1.
6. Repeat steps 4 a–d to wash PIPs a **third** time. Use 350 µl 0.5X PWB1.
   Process two samples at a time to ensure optimal PIP pelleting and to avoid PIP loss.
7. Centrifuge for 1 minute.
8. While the tube is still at an angle, immediately remove and discard supernatant to the 0.1 ml mark on the tube.
9. Store samples on ice and proceed immediately to cDNA amplification.

Amplify cDNA

Consumables

- PMM4 (4X PCR Master Mix)
- WTAP (WTA Primer)
- 0.2 ml 8-tube strips and caps

Preparation

1. Prepare the following consumables:
   - PMM4—Pipette to mix, and then centrifuge briefly.
   - WTAP—Pipette to mix, and then centrifuge briefly.
2. Save the following cDNA Amplification program on the thermal cycler:
   - Set the lid temperature to 105°C
   - 95°C for 3 minutes
   - 5 cycles of:
     - 98°C for 15 seconds
     - 69°C for 10 minutes
   - 72°C for 5 minutes
   - Hold at 4°C

   Annealing and extension both occur at 69°C in this profile.

   The specified program parameters are critical for assay performance. Do not modify the PCR cycle number. Altering the program causes unusable results.

Procedure

1. Combine the following volumes to prepare Whole Transcriptome Amplification (WTA) Master Mix:

Reagent	Volume Per Reaction (µl)	4.4X Reaction Master Mix (µl)	8.8X Reaction Master Mix (µl)
PMM4	33.8	149	297
WTAP	1.4	6	12
Total	35.2	155	309

2. Vortex to mix, and then centrifuge briefly.
3. Add 35 µl WTA Master Mix to each sample tube.
4. Flick to mix, and then pulse vortex for 5 seconds.

5. Distribute the WTA reaction mixture into 2 x 68 µl aliquots in an 8-tube strip. Centrifuge the WTA reaction mixture briefly before dispensing the last aliquot, and then transfer all of the remaining volume.
6. Label the 8-tube strips appropriately so each sample can be recombined after cDNA amplification.
7. Place on the preprogrammed thermal cycler and run the cDNA Amplification program.

SAFE STOPPING POINT

If you are stopping, leave the sample tubes on the thermal cycler at 4°C or store samples at 2°C to 8°C overnight.

Prepare for Protocol

This preparation is required to perform the protocol steps leading up to the next safe stopping point.

1. Remove the reagents from the box and thaw as follows.

Table 18 2°C to 8°C Storage

Reagent	Box Name	Thaw Instructions
CEB	Illumina Single Cell 3′ RNA Capture, T10, Box 2	Use at room temperature.
IPB2	Illumina Single Cell 3′ RNA Capture, T10, Box 2	Let stand for 30 minutes to bring to room temperature.
PWB1	Illumina Single Cell 3′ RNA Capture, T10, Box 2	Keep on ice.

Table 19 -25°C to -15°C Storage

Reagent	Box Name	Thaw Instructions
PMM4	Illumina Single Cell 3′ RNA Capture, T10, Box 3	Keep on ice.
WTAP	Illumina Single Cell 3′ RNA Capture T10, Box 3	Thaw on ice.

Isolate cDNA from PIPs

Consumables

- CEB (cDNA Extraction Buffer)
- 1.5 ml tubes (emulsion safe)

Preparation

1. Label one new 1.5 ml tube per sample.

Procedure

1. If you are resuming after a safe stopping point, briefly centrifuge the tubes.
2. Add 100 µl CEB to each cDNA reaction in the 8-tube strips.
3. Transfer the 2 aliquots of each sample to a new emulsion-safe 1.5 ml tube per sample.
4. Centrifuge the 8-tube strips briefly, and then transfer any remaining beads and supernatant to the appropriate 1.5 ml sample tube.
5. Pulse vortex, and then centrifuge for 1 minute.
6. Without disturbing the pellet, transfer 200 µl supernatant to a new 1.5 ml tube per sample.
7. Add 200 µl CEB to the 1.5 ml tube containing the PIP pellet.
8. Pulse vortex, and then centrifuge for 1 minute.
9. Without disturbing the pellet, transfer 200 µl supernatant from the tube containing the PIP pellet to the tube containing the transferred supernatant.
   The total volume of supernatant per sample is 400 µl.
10. Briefly centrifuge the tube containing the supernatant, and then examine the bottom of the tubes for PIPs.
11. If residual PIPs pellets are present, transfer the supernatant to a new 1.5 ml tube, and then measure the transferred supernatant volume.
    If the final volume of supernatant is not 400 µl per sample, adjust the magnetic bead volume during magnetic bead purification.
12. Save the remaining PIP pellet and store at -85°C to -65°C for up to 3 months.

> The saved pellets are required for Illumina Single Cell Supplemental Enrichment and Amplification Kit applications. If library preparation fails to produce sequencing-quality libraries, the saved pellets can be used to recover cDNA.

Clean Up cDNA

Consumables

- IPB2 (Illumina Purification Beads 2)
- IDTE Buffer
- Absolute ethanol (EtOH)
- Nuclease-free water (customer provided)

For Research Use Only. Not for use in diagnostic procedures.

Preparation

1. Prepare the following consumables:
   - IPB2—Vortex to resuspend.
   - For each reaction, prepare 2 ml fresh 85% EtOH solution from absolute EtOH.

Procedure

1. For each 400 µl reaction volume, add 320 µl IPB2. If needed, adjust the magnetic bead volume for a 0.8X ratio of IPB2.
2. Vortex until thoroughly mixed and homogenous, then pulse centrifuge to bring the liquid to the bottom of the tube.
3. Incubate at room temperature for 5 minutes.
4. Place on the magnetic stand and wait until the liquid is clear (~5 minutes).
   If magnetic beads do not pellet after 5 minutes, there might be PIP carryover. Leave on the magnetic stand until liquid is clear.
5. Without disturbing the beads, remove and discard supernatant.
6. Wash beads as follows.
   a. Add 1 ml fresh 85% EtOH.
   b. Wait 30 seconds.
   c. Without disturbing the pellet, remove and discard 1 ml EtOH.
7. Wash the beads a **second** time.
8. Without disturbing the beads, use a 20 µl pipette to remove residual EtOH.
9. Air-dry until a slight gloss remains on the bead surface (~2 minutes).
   Depending on humidity level, air-drying can take up to 5 minutes.

   Overdrying the beads can result in cracks and decrease elution efficiency.
10. Remove from the magnetic stand.
11. Add 42 µl IDTE Buffer.
12. Pipette to resuspend.
13. Pulse centrifuge to bring the liquid to the bottom of the tube.
14. Incubate at room temperature for 5 minutes.
    If bead pellets are overdried, pipette mix frequently during incubation to minimize sample loss.
15. Place into the magnetic stand and wait until the mixture is clear (~2 minutes). Keep the 1.5 ml tubes on the magnetic stand.
16. Transfer 40 µl supernatant to a new PCR tube strip and place the strip immediately on ice.

For Research Use Only. Not for use in diagnostic procedures.

This cDNA is used as input for library preparation. The remaining 2 µl supernatant on the magnetic stand is used for QC.

17. Store the cDNA for library preparation at -85°C to -65°C for up to 72 hours.

> Storing cDNA in a high amplicon-contaminated area might affect the quality of your samples. Storing cDNA for longer than 72 hours can result in final library QC failure from degraded cDNA.

Assess cDNA Quality

This step uses the remaining elution volume (2 µl) from the magnetic bead pellet to assess cDNA QC.

Consumables

- PMM4 (4X PCR Master Mix)
- WTAP (WTA Primer)
- PWB1 (Wash Buffer)
- Nuclease-free water (customer provided)
- 0.2 ml 8-tube strips

Preparation

1. Prepare the following consumables:
   - 0.5X PWB1—Multiply each of the following volumes by the number of samples.
     - PWB1 (7 µl)
     - Nuclease-free water (7 µl)

     These volumes produce 14 µl 0.5X PWB1 per sample.
   - PMM4—Pipette to mix, and then centrifuge briefly.
   - WTAP—Pipette to mix, and then centrifuge briefly.
2. Save the following QC Amplification program on the thermal cycler:
   - Set the lid temperature to 105°C
   - 95°C for 3 minutes
   - X cycles of:
     - 98°C for 15 seconds
     - 69°C for 4 minutes
   - 72°C for 5 minutes
   - Hold at 4°C

Adjust the program to optimize for different input quality and concentration.

Annealing and extension both occur at 69°C in this profile.

Sample Input	Number of QC PCR Cycles (X)
High RNA sample types (eg, cell lines)	9
Low RNA sample types (eg, primary cells and nuclei)	12

Procedure

1. Keep on the magnetic stand and add 9 µl nuclease-free water.
2. Without disturbing the pellet, pipette to mix.
3. Incubate at room temperature for 1 minute.
4. Without disturbing the beads, transfer 10 µl from the sample tube to a new PCR tube strip.
5. Place the sample on ice.
6. Combine the following volumes to prepare QC Master Mix:

Reagent	Volume Per Reaction (µl)	4.4X Reaction Master Mix (µl)	8.8X Reaction Master Mix (µl)
PMM4	6.25	27.5	55
WTAP	1	4.4	8.8
0.5X PWB1	7.75	34.1	68.2
Total	15	66	132

7. Vortex to mix, and then centrifuge briefly.
8. Add 15 µl QC Master Mix to each sample.
   The total volume per sample is 25 µl.
9. Pulse vortex and then centrifuge briefly.
10. Place on the preprogrammed thermal cycler and run the QC Amplification program.

SAFE STOPPING POINT

If you are stopping, leave the samples on the thermal cycler at 4°C or store samples at 2°C to 8°C overnight.

Prepare for Protocol

This preparation is required to perform the protocol steps leading up to the next safe stopping point.

1. Remove the reagents from the box and thaw as follows.

Table 20 2°C to 8°C Storage

Reagent	Box Name	Thaw Instructions
IPB2	Illumina Single Cell 3′ RNA Capture, T10, Box 2	Let stand for 30 minutes to bring to room temperature.

Clean Up QC Product

Consumables

- IPB2 (Illumina Purification Beads 2)
- IDTE Buffer
- Absolute ethanol (EtOH)
- Nuclease-free water (customer provided)
- 0.2 ml 8-tube strips

Preparation

1. Prepare the following consumables:
   - IPB2—Vortex to resuspend.
2. For each reaction, prepare 400 µl fresh 85% EtOH solution from absolute EtOH.

Procedure

1. Add 15 µl nuclease-free water to each sample.
   The total volume is 40 µl per sample.
2. Add 32 µl IPB2 for a 0.8X ratio.
3. Pipette 15 times at the 67 µl stroke.
4. Incubate at room temperature for 5 minutes.
5. Place on the magnetic stand and wait until the liquid is clear (~5 minutes).
6. Without disturbing the pellet, remove and discard all supernatant.
7. Wash beads as follows.
   a. Add 200 µl fresh 85% EtOH.
   b. Wait 30 seconds.
   c. Without disturbing the beads, remove and discard supernatant.
8. Wash the beads a **second** time.
9. Without disturbing the beads, use a 20 µl pipette to remove residual EtOH.
10. Air-dry until a slight gloss remains on the bead surface (~2 minutes).

For Research Use Only. Not for use in diagnostic procedures.

Depending on humidity level, air-drying can take up to 5 minutes.

Overdrying the beads can result in cracks and decrease elution efficiency.

11. Remove from the magnetic stand.
12. Add 11 µl IDTE Buffer over the bead pellet in the tube.
13. Pipette to resuspend.
14. If necessary, briefly centrifuge to bring the liquid to the bottom of the tube.
15. Incubate at room temperature for 5 minutes.
    If bead pellets are overdried, pipette mix frequently during incubation to minimize sample loss.
16. Place into the magnetic stand and wait until the mixture is clear (~2 minutes).
17. Transfer 10 µl supernatant from the sample tube to a new PCR tube.

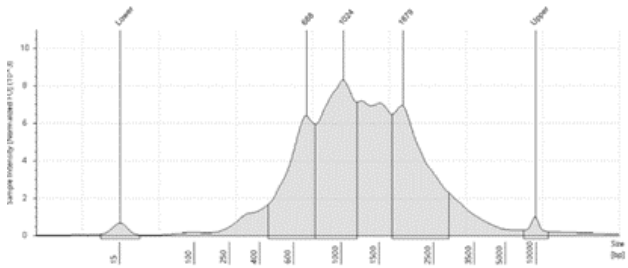
Check QC Product

Perform the following to check the quantity and quality of the QC product.

The QC product is stable at -25°C to -15°C for up to four weeks. If necessary, you can submit the QC product with the final libraries when using a core lab for fragment analysis.

1. Quantify 2 µl of each sample using a Qubit High Sensitivity kit.
2. Assess average fragment size using an Agilent Bioanalyzer or Tapestation.
   - If necessary, dilute samples to make sure that they are within the appropriate range for the device.
   - Load an appropriate volume of the purified library on the fragment analyzer.
     Recommended loading concentration is 1–10 ng/µl on Tapestation or 1–2 ng/µl on Bioanalyzer.

   Loading less than 1 ng/µl may not give accurate results.
3. Proceed to library preparation upon successful cDNA QC. The average fragment size for cDNA should be > 500 bp as determined by the fragment analyzer region table (set for 200–5000 bp). Average size may vary depending on cell type and experimental condition.



Representative cDNA traces for a 15,000 cell input human/mouse cell mixture (HEK 293T/NIH 3T3) using a high sensitivity D5000 ScreenTape.

Prepare for Protocol

This preparation is required to perform the protocol steps leading up to the next safe stopping point.

1. If you are resuming after a safe stopping point, remove cDNA from storage and place on ice.

   It is critical to use the total 40 µl volume of cDNA recovered during *Clean Up cDNA* on page 34 for library preparation. Using the QC product as input into library preparation results in assay failure.

2. Remove the reagents from the box and thaw as follows.

   Table 21 2°C to 8°C Storage

Reagent	Box Name	Thaw Instructions
IPB2	Illumina Single Cell Library Prep, Box 1	Let stand for 30 minutes to bring to room temperature.
NFW	Illumina Single Cell Library Prep, Box 1	Use at room temperature.

   Table 22 -25°C to -15°C Storage

Reagent	Box Name	Thaw Instructions
PMM4	Illumina Single Cell Library Prep, Box 2	Keep on ice.
LAM	Illumina Single Cell Library Prep, Box 2	Thaw on ice.
LPB	Illumina Single Cell Library Prep, Box 2	Thaw on ice.
LPE	Illumina Single Cell Library Prep, Box 2	Thaw on ice.
LPMA	Illumina Single Cell Library Prep, Box 2	Thaw on ice.

3. Remove one of the following index adapter options from storage and thaw as follows.

   Table 23 -25°C to -15°C Storage

Reagent	Box Name	Thaw Instructions
Illumina Single Cell Prep UDI Mix Strip	Illumina Single Cell Library Prep, Box 2	Thaw on ice.
Illumina Single Cell Unique Dual Indexes plate	Illumina Single Cell Unique Dual Indexes (96 Indexes, 96 Samples)	Thaw on ice.

Fragmentation, End Repair, & A-Tailing

Consumables

- LPB (Library Prep Buffer)
- LPE (Library Prep Enzymes)

Preparation

1. Prepare the following consumables:
   - LPB—Vortex to mix, and then centrifuge briefly. Keep on ice.
   - LPE—Pipette to mix, and then centrifuge briefly. Keep on ice.
2. Save the following cDNA Fragmentation program on the thermal cycler:
   - Set the lid temperature to 105°C
   - Hold at 4°C
   - 30°C for 6 minutes
   - 65°C for 30 minutes
   - Hold at 4°C
3. Run the cDNA Fragmentation program to prechill the thermal cycler block to 4°C and preheat the lid to 105°C.

Procedure

1. Combine the following volumes to prepare Library Prep Master Mix.

Reagent	Volume Per Reaction (µl)	4.4X Reaction Master Mix (µl)	8.8X Reaction Master Mix (µl)
LPB	4	17.6	35.2
LPE	6	26.4	52.8
Total	10	44	88

2. Vortex to mix, and then centrifuge briefly.
3. Add 10 µl Library Prep Master Mix to each tube containing 40 µl of cDNA.
   The total volume is 50 µl.
4. Vortex to mix or pipette mix 10 times at the 25 µl stroke.
   Make sure to mix completely.
5. Centrifuge briefly. Place the samples on ice.

6. Place on the prechilled thermal cycler, and then use the skip step function to run the next steps of the cDNA Fragmentation program.
7. When the thermal cycler program reaches 4°C, immediately remove the sample and proceed to the next step.

Ligate Adapters

Consumables

- LAM (Library Adapter Mix)
- LPMA (Library Prep Mix A)
- NFW (Nuclease-Free Water) (kit provided)

Preparation

1. Prepare the following consumables:
   - LPMA—Pipette slowly to mix, and then centrifuge briefly. Keep on ice.
2. Save the following Ligation program on the thermal cycler:
   - 20°C for 20 minutes
   - Ensure that the heated lid is off.

Procedure

1. Combine the following volumes to prepare diluted LAM. For multiple samples, multiply each volume by the number of samples plus 10% overage.
   - LAM (0.75 µl)
   - NFW (4.25 µl)

   These volumes produce 5 µl per sample.

   Store dilutions of the LAM for up to one week at -25°C to -15°C.
2. Add 5 µl diluted LAM to each reaction.
3. Vortex to mix, and then centrifuge briefly.
4. Add 20 µl LPMA to each reaction.
   LPMA is viscous, pipette slowly to ensure that proper volumes are dispensed. The total volume per reaction is 75 µl.
5. Pipette 10 times at 40 µl stroke, and then briefly centrifuge.
6. Place in the thermal cycler and run the Ligation program.

Clean Up Ligated Fragments

This step uses magnetic beads to purify the adapter-ligated fragments.

Consumables

- IPB2 (Illumina Purification Beads 2)
- Absolute ethanol (EtOH)
- NFW (Nuclease-Free Water) (kit provided)
- 0.2 ml 8-tube strip

Preparation

1. Prepare the following consumables:
   - IPB2—Vortex to resuspend.
   - For each reaction, prepare 400 µl fresh 85% EtOH solution from absolute EtOH.

Procedure

1. Add 60 µl IPB2 to each sample.
2. Pipette 15 times at the 125 µl stroke to mix.
3. Incubate at room temperature for 5 minutes.
4. Place on the magnetic stand and wait until the liquid is clear (~5 minutes).
5. Without disturbing the beads, remove and discard all supernatant from each tube.
6. Wash beads as follows.
   a. Add 200 µl fresh 85% EtOH.
   b. Wait 30 seconds.
   c. Without disturbing the pellet, remove and all supernatant.
7. Wash the beads a **second** time.
8. Without disturbing the beads, use a 20 µl pipette to remove residual EtOH.
9. Air-dry until a slight gloss remains on the bead surface (~2 minutes).
   Depending on humidity level, air-drying can take up to 5 minutes.

   Overdrying the beads can result in cracks and decrease elution efficiency.
10. Remove from the magnetic stand.
11. Add 34 µl nuclease-free water over the bead pellet in the tube.
12. Pipette 10 times at the 32.5 µl stroke to resuspend.
13. Incubate at room temperature for 5 minutes.
14. Place on the magnetic stand and wait until the liquid is clear (~2 minutes).
15. Transfer 32.5 µl supernatant to a new 8-tube strip.

Amplify Library

Choose and record a unique index mix for each sample to ensure that no sample index combinations overlap in a multiplexed sequencing run.

Align the orientation of the hole on the side of the strip containing the indexes with the index map in the box lid and in the following diagram:



Pierce the foil with a 20 µl pipette before aspirating. Check orientation of index tube strip to ensure that the correct index combination is added and recorded for sequencing. There are eight index mixes provided with this kit to allow for unique dual indexing of eight samples.

To pool more than eight libraries together in a sequencing run, use the Illumina Single Cell Unique Dual Indexes plate. For more information on pooling, refer to *Index Adapters Pooling Guide*. For more information on recommended indexes, refer to *Illumina Adapter Sequences*.

Consumables

- PMM4 (4X PCR Master Mix)
- One of the following index set options:
  - Illumina Single Cell Prep UDI Mix Strip
  - Illumina Single Cell Unique Dual Indexes plate

Preparation

1. Prepare the following consumables:
   - PMM4—Pipette 10 times, and then centrifuge briefly.
2. Save the following PCR program on the thermal cycler:
   - Set the lid temperature to 105°C
   - 98°C for 45 seconds
   - X cycles of:
     - 98°C for 15 seconds
     - 67°C for 30 seconds
     - 69°C for 45 seconds
   - 72°C for 1 minute
   - Hold at 4°C

   Adjust the program to optimize for different input quality and concentration.

Sample Input	Number of PCR Cycles (X)
High RNA sample types (eg, cell lines)	10
Low RNA sample types (eg, primary cells and nuclei)	15

Decrease by 1–2 cycles for cDNA QC concentrations > 200 ng/µl.

Procedure

1. Add the following volumes to each reaction tube, ***in the order listed***:
   - Unique Library Index Mix (5 µl)
   - PMM4 (12.5 µl)

   These volumes produce 50 µl PCR mix per tube.
2. Pipette mix 10 times at 32 µl stroke, and then briefly centrifuge.
3. Place on the preprogrammed thermal cycler and run the PCR program.

SAFE STOPPING POINT

If you are stopping, leave the samples on the thermal cycler at 4°C or store samples at 2°C to 8°C overnight.

Prepare for Protocol

This preparation is required to perform the protocol steps leading up to the next safe stopping point.

1. Remove the reagents from the box and thaw as follows.

   Table 24 2°C to 8°C Storage

Reagent	Box Name	Thaw Instructions
IPB2	Illumina Single Cell Library Prep, Box 1	Let stand for 30 minutes to bring to room temperature.

Clean Up Library

Consumables

- IPB2 (Illumina Purification Beads 2)
- IDTE Buffer
- Absolute ethanol (EtOH)
- Nuclease-free water (customer provided, sterile)
- 0.2 ml 8-tube strip

Preparation

1. Prepare the following consumables:
   - IPB2—Vortex to resuspend.
   - For each reaction, prepare 400 µl fresh 85% EtOH solution from absolute EtOH.

Procedure

1. If you are resuming after a safe stopping point, briefly centrifuge the sample tube.
2. Add 45 µl nuclease-free water.
3. For 95 µl reaction volumes, add 76 µl IPB2. If needed, adjust the magnetic bead volume for a 0.8X ratio of magnetic beads.
4. Pipette 15 times at the 160 µl stroke to mix.
5. If any magnetic beads remain on the lid or side of the tube, centrifuge briefly.
6. Incubate at room temperature for 5 minutes.
7. Place on the magnetic stand and wait until the liquid is clear (~5 minutes).
8. Without disturbing the beads, remove and discard all supernatant from each tube.
9. Wash beads as follows.
   a. Add 200 µl fresh 85% EtOH.
   b. Wait 30 seconds.
   c. Without disturbing the pellet, remove and discard EtOH.

For Research Use Only. Not for use in diagnostic procedures.

10. Wash the beads a **second** time.
11. Without disturbing the beads, use a 20 µl pipette to remove residual EtOH.
12. Air-dry until a slight gloss remains on the bead surface (~2 minutes).
    Depending on humidity level, air-drying can take up to 5 minutes.

    Overdrying the beads can result in cracks and decrease elution efficiency.
13. Remove from the magnetic stand.
14. Add 21 µl IDTE Buffer.
15. Pipette 10 times at the 20 µl stroke to resuspend.
16. Incubate at room temperature for 5 minutes.
17. Place on the magnetic stand and wait until the liquid is clear (~2 minutes).
18. Transfer 20 µl supernatant to a new PCR tube strip.

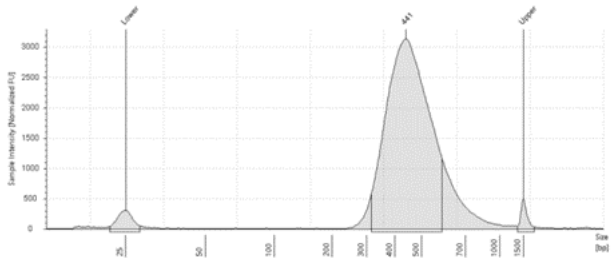
SAFE STOPPING POINT

If you are stopping, seal the tubes and store at -25°C to -15°C for up to 30 days.

Check Library

This step checks the concentration and quality of the final libraries.

1. Quantify 2 µl of each sample using a Qubit High Sensitivity kit.
2. Assess average fragment size using an Agilent Bioanalyzer or Tapestation.
   - If necessary, dilute samples to make sure that they are within the appropriate range for the device.
   - Load an appropriate volume of the purified library on the fragment analyzer.
     Recommended loading concentration is 1–10 ng/µl on Tapestation or 1–2 ng/µl on Bioanalyzer.



Representative library trace (expected avg. size 370–550 bp) for a 10,000 cell capture PBMC cell mixture on a High Sensitivity D1000 ScreenTape.

For Research Use Only. Not for use in diagnostic procedures.

Dilute Libraries to the Starting Concentration

This step dilutes libraries to the starting concentration for your sequencing system and is the first step in a serial dilution. After diluting to the starting concentration, libraries are ready to be denatured and diluted to the final loading concentration.

1. Calculate the molarity value of the libraries using the following formula:

$$\frac{ng/\mu l \times 10^6}{660\frac{g}{mol} \times average\ library\ size\ (bp)} = Molarity\ (nM)$$

2. Using the molarity value, calculate the volumes of RSB and library needed to dilute libraries to the starting concentration for your system

Sequencing System	Starting Concentration (nM)	Final Loading Concentration (pM)	Recommended PhiX Spike	Denature and Dilute Protocol
NextSeq 500/550	2	1.6	1%	Illumina Single Cell 3′ RNA Prep
NextSeq 1000/2000	2	550	1%	Onboard or XLEAP-SBS Onboard
NovaSeq 6000	1	200	1%	Standard
NovaSeq X Series	2	200	2%	Standard

3. Dilute each library to the starting concentration for your system and then pool equal volumes of each library to create a multiplexed library pool.
4. Follow the denature and dilute instructions for your system in the Denature and Dilute protocol generator.

For Research Use Only. Not for use in diagnostic procedures.

Resources & References

The Illumina Single Cell 3′ RNA Prep pages on the Illumina support site provide additional resources. These resources include training, compatible products, and other considerations. Always check the support pages for the latest versions.

Revision History

Document	Date	Description of Change
Document # 200063182 v01	August 2025	Added instructions for diluting libraries to starting concentrations and relevant sequencing information. Added alternative dry bath and starter equipment kit options. Corrected volume of 0.5X Wash Buffer to prepare for assessing cDNA quality. Corrected volume of ethanol to prepare for cDNA clean up. Updated vortexer setting guidance. Updated catalog number for user-supplied USA Scientific 0.2 ml PCR 8-tube strips and caps. Updated catalog number for user-supplied Fisher Scientific 0.5 ml PCR Safe-Lock PCR Clean tubes.
Document # 200063182 v00	March 2025	Initial release.

For Research Use Only. Not for use in diagnostic procedures.

Illumina, Inc.
5200 Illumina Way
San Diego, California 92122 U.S.A.
+1.800.809.ILMN (4566)
+1.858.202.4566 (outside North America)
techsupport@illumina.com
www.illumina.com

For Research Use Only. Not for use in diagnostic procedures.

© 2025 Illumina, Inc. All rights reserved.

illumina®